TECHNICAL ARCHITECTURE & DATABASE DESIGN

Entity-Relationship Diagram & Data Model Specification

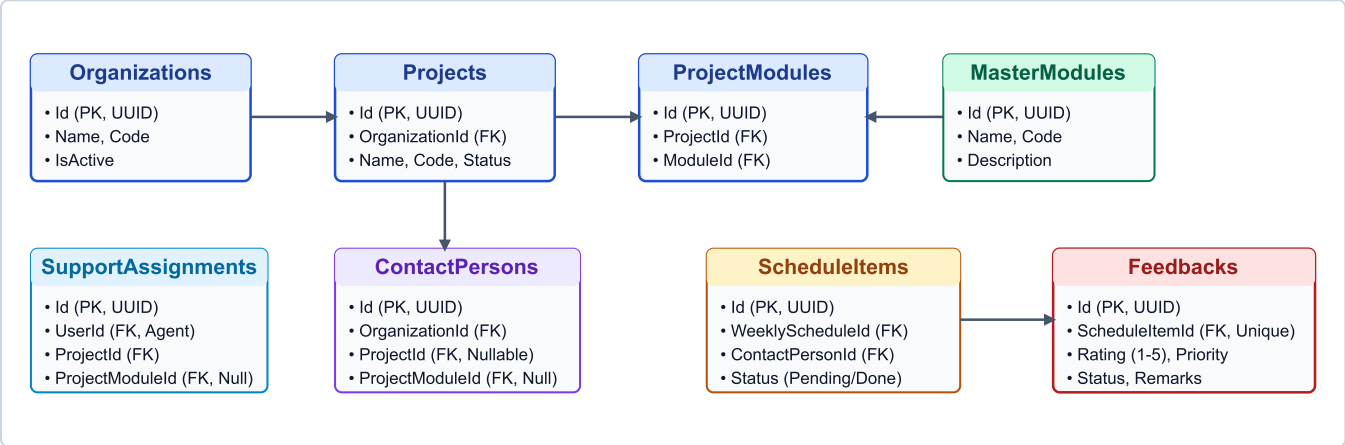
Relational Structure: Organizations > Projects > Project Modules > Contact Hierarchy

1. Architecture Overview & Relational Hierarchy

To satisfy multi-tenant client requirements, the schema strictly decouples **Organizations** (Customers), **Projects**, and **Modules** into a normalized 3-tier hierarchy:

- **Organization (Customer):** The parent commercial entity (e.g., Bank, Ministry, Enterprise).
- **Project:** Independent software deliverables under an Organization (e.g., Core Banking Upgrade, Mobile Portal).
- **ProjectModule:** Granular packages/components mapped to a project (e.g., Loans, Audit Engine, Biometrics).
- **ContactPerson:** Contacts can be bound at the Organization level (General Director), Project level (Project Manager), or Module level (Lead Specialist).

2. Visual Entity-Relationship Diagram (ERD)



3. Core Model Definitions & Schema Tables

3.1 Client Hierarchy & Master Data

Table / Field	Data Type	Constraints	Description & Relationship
Organizations			Represents the paying customer entity.
`Id`	`uniqueidentifier`	PK	Unique primary identifier.
`Name`, `Code`	`nvarchar(150/50)`	Unique Index	Legal company name and alphanumeric code.
Projects			Distinct systems/engagements under an Organization.
`Id`	`uniqueidentifier`	PK	Unique project identifier.
`OrganizationId`	`uniqueidentifier`	FK Index	Parent customer reference (`ON DELETE RESTRICT`).
`Name`, `ProjectStatus`	`nvarchar(150)`, `int`	Required	Project Title & lifecycle state (Active, Maintenance).
ProjectModules			Specific operational module enabled for a given project.
`Id`	`uniqueidentifier`	PK	Unique instance mapping identifier.
`ProjectId`	`uniqueidentifier`	FK	Links module deployment to specific project.
`MasterModuleId`	`uniqueidentifier`	FK	Links to catalog catalog definition table.

3.2 Contacts & Workload Assignment Hierarchy

Table / Field	Data Type	Constraints	Description & Relationship
ContactPersons			Hierarchical contact registry per org, project, or module.
`Id`	`uniqueidentifier`	PK	Primary key.
`OrganizationId`	`uniqueidentifier`	FK	Direct parent organization relationship.
`ProjectId`	`uniqueidentifier`	FK (Nullable)	Optional binding: specific project manager/sponsor.
`ProjectModuleId`	`uniqueidentifier`	FK (Nullable)	Optional binding: specific operational module focal point.
`FullName`, `Phone`	`nvarchar(120/30)`	Required	Caller verification contact points.
`IsPrimary`	`bit`	Default 0	Fallback priority when generating scheduled calls.
SupportAssignments			Workload dispatch linking agents to projects or modules.
`UserId`, `ProjectId`	`uniqueidentifier`	FK	Agent assigned to oversee project customer success.
`ProjectModuleId`	`uniqueidentifier`	FK (Nullable)	Optional: restrict agent to a single specialized module.

3.3 Scheduling, Feedback & Performance Evaluation

Table / Field	Data Type	Constraints	Description & Relationship
ScheduleItems			Individual weekly planned customer touchpoints.
`Id`	`uniqueidentifier`	PK	Schedule item primary key.
`WeeklyScheduleId`	`uniqueidentifier`	FK	Belongs to Monday–Sunday planning cycle.
`AssignedUserId`	`uniqueidentifier`	FK Index	Agent responsible for conducting the interview.
`ContactPersonId`	`uniqueidentifier`	FK	Pre-resolved targeted contact phone/person.
`Status`	`int (Enum)`	Index	`Pending = 0`, `Completed = 1`, `Skipped = 2`.
Feedbacks			Immutable audit record of the collected feedback.
`ScheduleItemId`	`uniqueidentifier`	FK Unique	Strict 1-to-1 link to planned touchpoint item.
`Rating`	`int`	Check (1–5)	CSAT quantitative satisfaction score.
`Priority`, `Status`	`int (Enums)`	Required	Priority (Low–Blocker) & Resolution status.
`Remarks`	`nvarchar(max)`	Required	Customer verbatim notes and action items.

Resolution Fallback Hierarchy for Schedule Generation: When compiling weekly planned calls, the generator targets: (1) `ProjectModule.ContactPerson` if an assignment is module-specific → (2) `Project.ContactPerson` (if designated) → (3) Fallback to `Organization.IsPrimary` contact person.